

MLOps / LLMOps

The system that ships.

CI/CD · Experiment Tracking · Model Registry · Observability

NCIIPC AI Grand Challenge

Alagu Prakalya, MLE @ ASAPP / Docker captain

91%

eval score

~~74%~~

production reality

Nobody knows why.

5 failure modes.

01 No reproducibility

same code · different results

02 No experiment comparison

you compare against memory

03 No path to production

notebook → service takes weeks

04 Training-serving skew

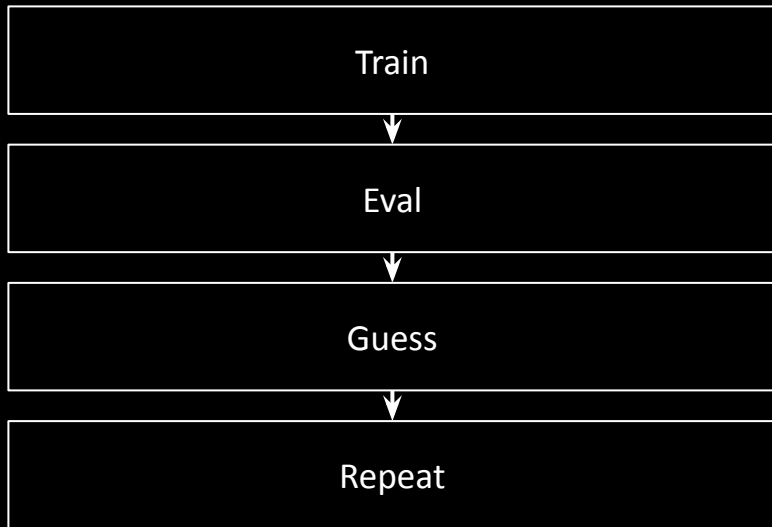
← we demo this

05 Silent model degradation

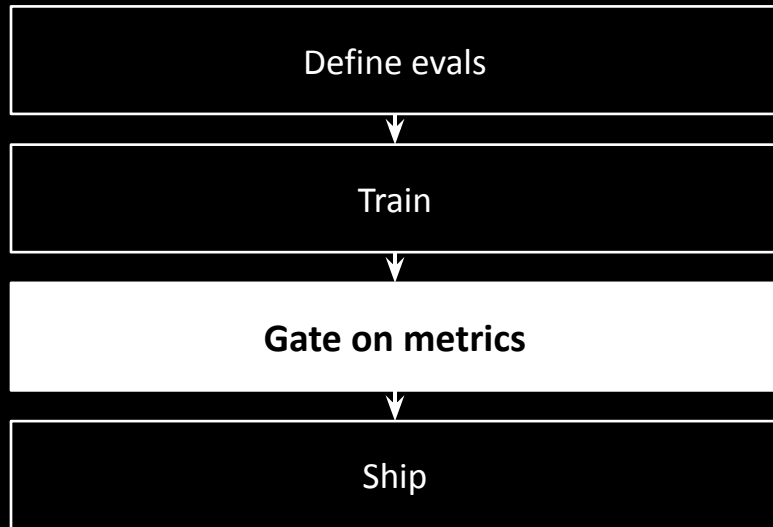
no alert until a user complains

Write evals first.

THE OLD WAY



THE RIGHT WAY



Four numbers. One contract.

Faithfulness

0.84

Claims in the answer supported
by retrieved context.
< 0.75 = hallucination.

Context Recall

0.79

Ground-truth info present in
retrieved passages.
Low = retriever failing, not the model.

Context Precision

0.70

Fraction of retrieved content
that is actually relevant.
Low = model drowning in noise.

Answer Relevancy

0.88

Does the answer address
the question asked?
Catches technically correct but useless.

Three lines.

```
1 run = wandb.init(  
2     project = "iit-delhi-demo",  
3     config  = config,  
4     tags    = ["qlora", "mistral-7b"],  
5 )  
6  
7 # loss curves · GPU mem · grad norms → automatic
```

Loss curves · GPU memory · Gradient norms · System metrics — all automatic.

One prompt edit.

↑ **0.23 faithfulness.**

v1 <i>Retired</i>	
faithfulness	0.61
context_recall	0.58
answer_relevancy	0.70
<i>Missing: 'Do not use prior knowledge'</i> <i>Missing: explicit I-don't-know path</i>	

4 lines
→ added →

v2 → production	
faithfulness	0.84
context_recall	0.79
answer_relevancy	0.88
<i>versioned · gated · audited</i>	

Cache the context.

90%

cost reduction

cache reads vs base price

85%

latency reduction

11.5s → 2.4s at 100K tokens

10×

cheaper reads

10% of base token price

Mechanism: K-V attention matrices computed once – reused on every cache hit. Static prompt prefix FIRST, user message LAST.

Clients resolve aliases, not UUIDs.

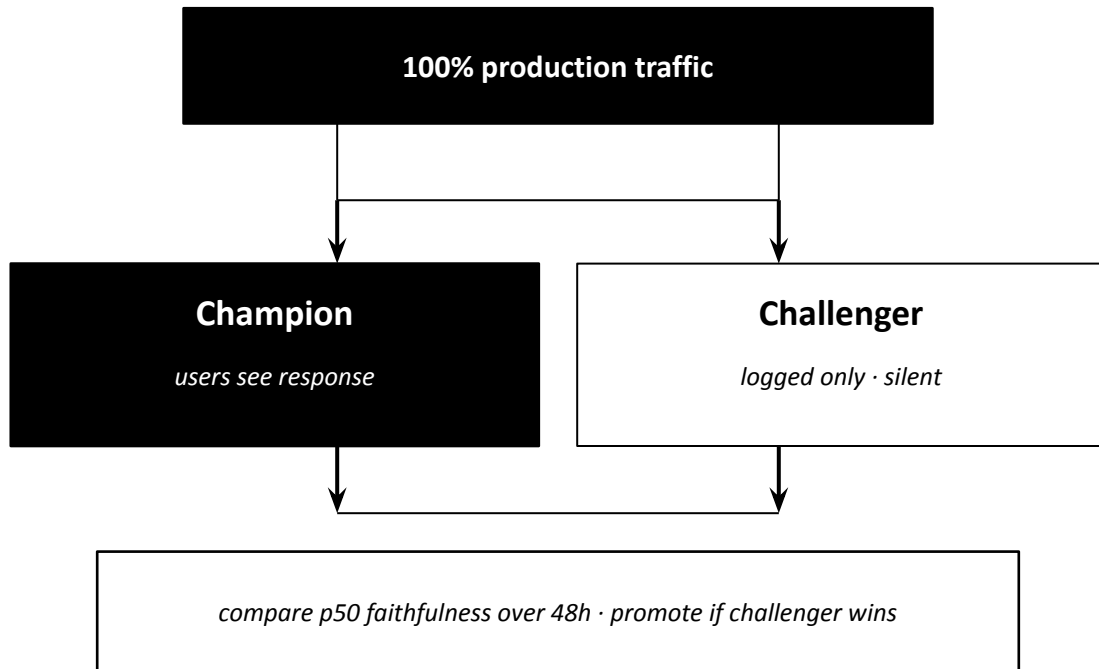
Zero client changes on every deploy.



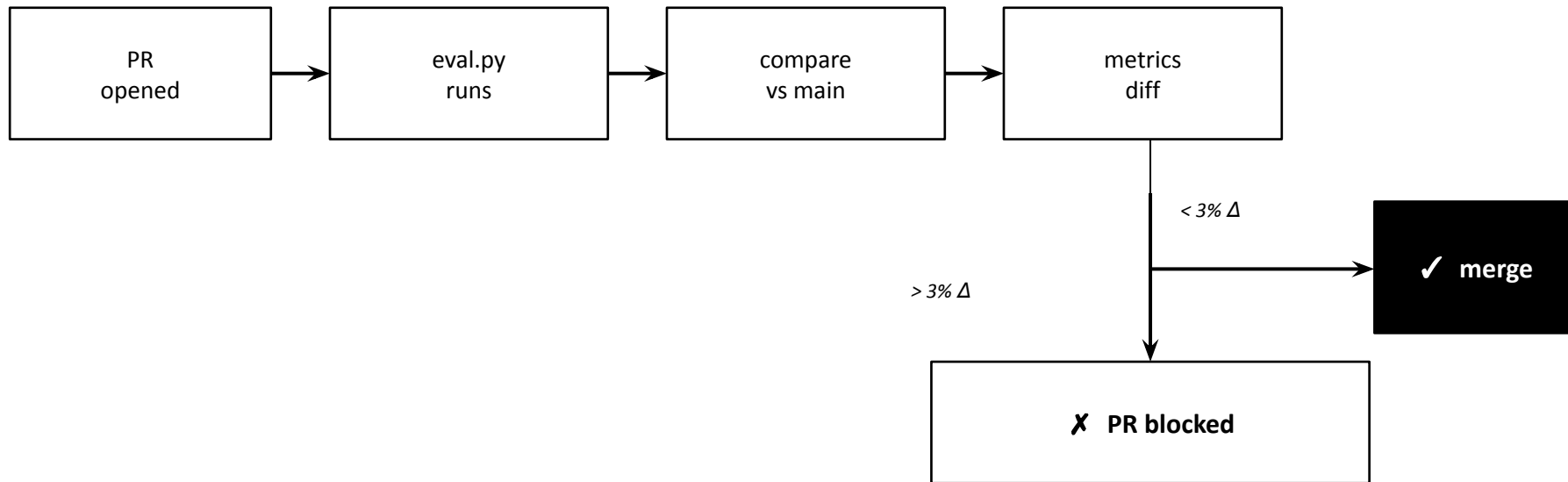
Promote a new model: update the alias. No redeployment. Full lineage preserved.

Ship nothing without comparison.

Netflix · Uber · LinkedIn.



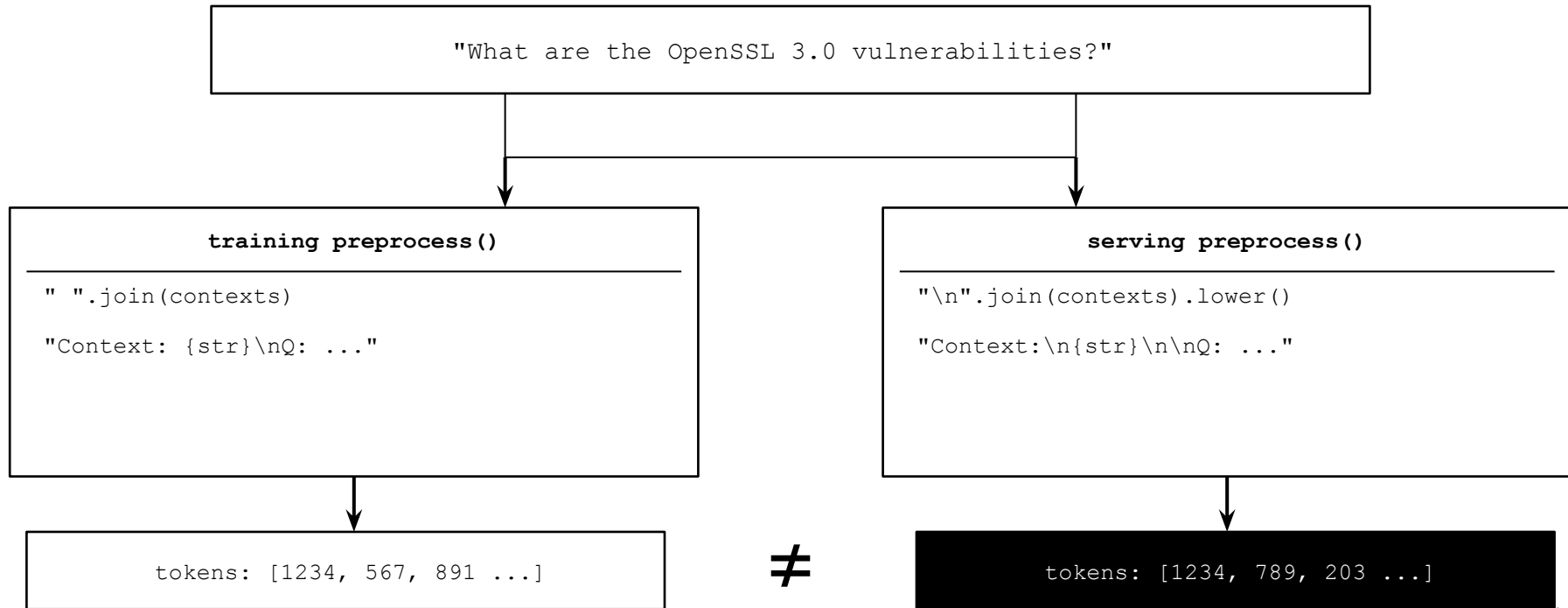
Regressions can't merge.



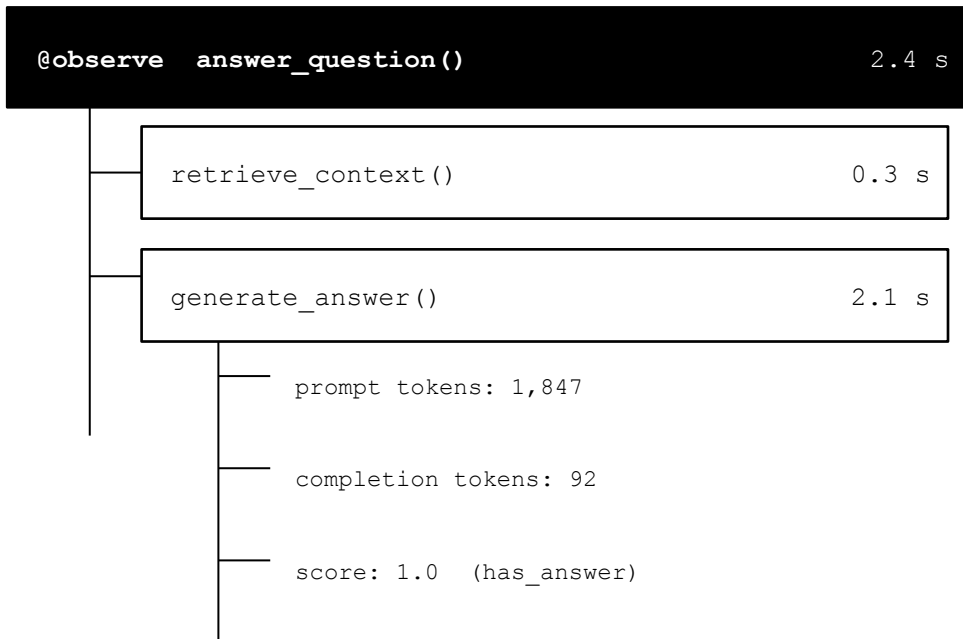
`sys.exit(1)` ← one line. GitHub enforces it.

Same input. Different tokens.

*No error raised.
Everything 'works'.*



W&B is for training. Langfuse is for prod.



Self-hosted

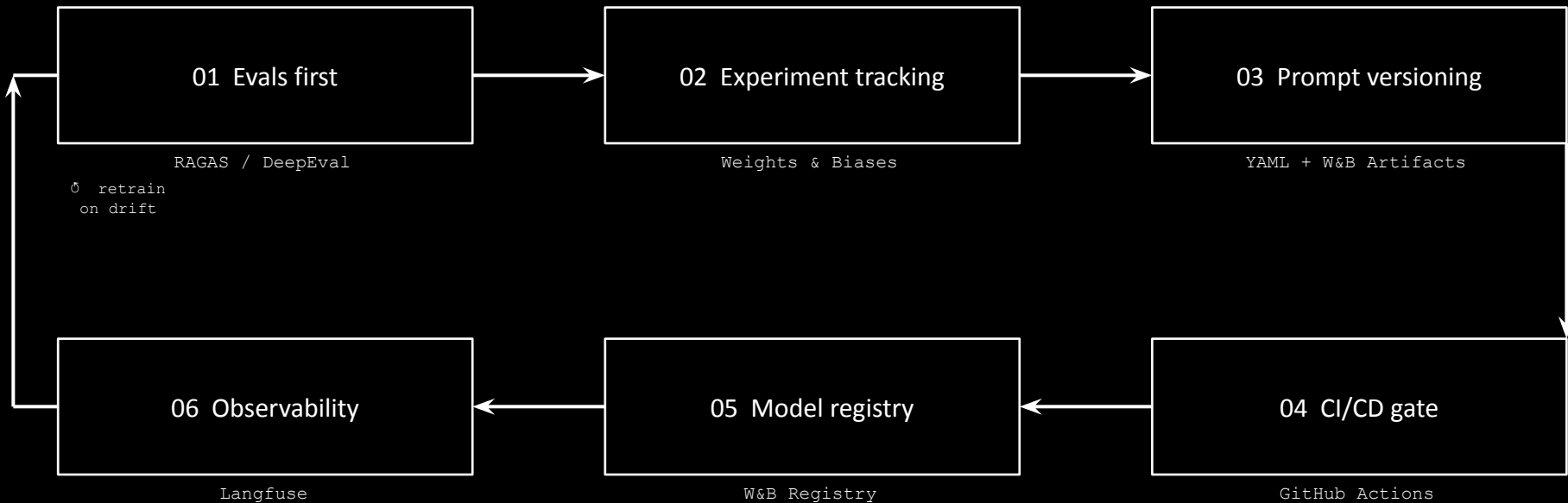
```
docker compose up -d langfuse
```

One decorator

@observe()

*No data leaves your VPC.
Critical for gov / defence deployments.*

From notebook to production.



make eval

Runs in 10 minutes on a laptop. RAGAS scores on your own questions.

`github.com/lagstill/mlops-demo`

Any Questions?



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